Beyond Metrics: A Human Validation Study of Procedural Personas in a Continuous Game Arena

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Abstract

Game-playing agents that embody distinct player archetypes (procedural personas) show promise for automated playtesting, yet virtually no work has empirically tested whether human observers actually perceive the intended behavioral styles. We present the first structured human validation of reinforcement-learning-trained procedural personas. Five persona archetypes are instantiated via reward shaping in a continuous, real-time 2D arena with multiple enemy types, treasures, and an exit goal. All agents are trained with Proximal Policy Optimization on documented consumer hardware. Evaluation combines quantitative gameplay metrics (kills, treasures, survival, speed) with a user study ($N = 34$) comprising forced-choice identification, Likert-scale consistency ratings, and selected believability items. The results, confirmed by statistical tests, show that the personas are quantitatively distinct, but perceptual recognition is uneven: extreme archetypes are reliably identified while subtler ones are confused. Importantly, believability did not correlate with recognition accuracy. This work bridges the human-validation gap in persona-based playtesting and provides a replicable, open-source baseline for future research.

Code — https://github.com/Re-G3X/ml-agents
Datasets — https://doi.org/10.5281/zenodo.21177978

1 Introduction

The growing complexity of modern video games imposes increasingly rigorous quality assurance demands. Developers must evaluate functional correctness alongside experiential qualities such as difficulty balancing, progression pacing (Hendrix et al. 2019), and the subjective perception of "fun" (Koster 2013), yet this evaluation still relies heavily on manual playtesting (Politowski, Petrillo, and Guéhéneuc 2021). Manual methods suffer from inherent limitations in scalability and consistency, which has motivated substantial research into automated alternatives (Albaghajati and Ahmed 2023; Politowski, Guéhéneuc, and Petrillo 2022).

Recent Reinforcement Learning (RL) breakthroughs have produced agents that surpass human performance in complex games (Vinyals et al. 2019; Wurman et al. 2022), highlighting their potential to transform video game testing. However, conventional game-playing agents typically pursue narrowly defined objectives like score maximization (Mnih et al. 2015), offering little variation in how game systems are exercised. In contrast, real players exhibit substantial strategic and motivational diversity (Tondello et al. 2019; Carneiro et al. 2022). This lack of behavioral diversity is a critical blind spot for automated testing. Without this diversity, automated agents explore only a constrained subset of game states, potentially overlooking critical bugs or balance issues that emerge under distinct play patterns (Holmgård et al. 2019).

The persona-based agent paradigm has emerged to address this. Rooted in established player-modeling research (Bartle 1996; Canossa and Drachen 2009), these agents represent distinct player archetypes, expressing characteristic decision-making tendencies (Ahlberg et al. 2023) or adapting priorities dynamically (Ariyurek, Surer, and Betin-Can 2023). By deploying a diverse set, developers can evaluate how different player types experience content, identifying a wider range of design issues (Guerrero-Romero, Lucas, and Perez-Liebana 2023) than non-persona-based approaches (Barthet et al. 2022).

Despite this promise, persona-based playtesting research has two major gaps. First, evaluation remains almost exclusively quantitative: studies report agreement ratios, trace distances, or outcome metrics, but no prior work has conducted a structured human evaluation of whether observers perceive the generated behaviors as consistent with the intended archetypes. Second, commonly used testbeds are predominantly turn-based grid worlds such as MiniDungeons 2 (Holmgård et al. 2016b), while broader benchmarks such as GVG-AI (Perez-Liebana et al. 2019) mostly feature discrete-action 2D games. While these environments support multi-objective play, our testbed expands persona evaluation into a continuous-space, real-time arena requiring 360° continuous motion alongside tactical evasion, item collection, and directional combat. Our study implements the original procedural personas within this hybrid continuous-action setting and evaluates them through both quantitative gameplay metrics and a structured human perception study.

We investigate a central research question: ***Can reward-shaped Deep Reinforcement Learning (DRL) agents produce personas that are quantitatively distinct and perceptually legible?*** This breaks into three sub-questions: **(SQ1)** Are gameplay profiles (kills, treasures, survival, speed) dis-

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tinguishable by archetype? **(SQ2)** Can observers identify intended archetypes above chance, and which are most confusable? **(SQ3)** Is perceived believability associated with recognition accuracy and distinctiveness? We treat believability as a complementary evaluation dimension rather than a prerequisite for automated testing: an agent may cover the intended play space yet appear mechanical, and measuring perceived naturalness reveals whether recognizability and human-like coherence pose distinct design goals.

We answer the questions through a two-stage evaluation. First, we train five personas using Proximal Policy Optimization (PPO) in a continuous arena with multiple enemies, treasures, and an exit. Second, a user study presents agent videos for forced-choice identification, Likert consistency ratings, and selected believability items (Guo, Adamo, and Mousas 2023). This provides the first structured human validation of procedural personas, directly testing whether quantitative behavioral differences translate into perceptible archetypes for human observers.

2 Related Work

Playtesting evaluates game experience aspects such as difficulty, balance, and enjoyment, yet manual playtesting methods suffer from fatigue, inconsistency, and limited coverage (Politowski, Petrillo, and Guéhéneuc 2021). These limitations have motivated automated playtesting methodologies that deliver continuous feedback (Albaghajati and Ahmed 2023). Recent RL breakthroughs have shown superhuman performance in complex games such as Gran Turismo Sport (Wurman et al. 2022), highlighting the potential of learning-based agents for testing. Within this context, persona-based approaches have emerged as a direct precursor to this current work (Holmgård et al. 2014a).

In game research, a persona denotes an archetype defined by consistent motivations and behaviors (Bartle 1996), constructed either as a designer-driven metaphor or as a data-driven lens from gameplay metrics (Canossa and Drachen 2009). Personas are distinguished from *playstyles*: the latter refer to tactical choices, while personas represent higher-order behavioral commitments expressed through multiple playstyles (Tychsen and Canossa 2008).

Holmgård et al. (2014b) presented *procedural personas* as generative player models, initially training Q-learning agents with tailored reward functions. They later replaced Q-learning with evolution strategies for better generalization (Holmgård et al. 2016a) and applied MCTS with evolved heuristics to MiniDungeons 2 (Holmgård et al. 2015, 2019). Subsequent work diversified methods: Iwasaki and Hasebe (2021; 2022) used NEAT with clustering; Ingram et al. (2023) proposed behavioural cloning from clustered human traces; Ferguson et al. (2022) used Dynamic Time Warping Imitation; Ahlberg et al. (2023) employed multimodal adversarial imitation learning; Ariyurek et al. (2023) introduced Alternate Path Finding with RL; and Fernandes et al. (2023; 2024) used evolutionary algorithms with emotionally-annotated playtraces.

Despite this diversity, a critical limitation persists: evaluation remains almost entirely quantitative. Studies report agreement ratios (Holmgård et al. 2016a), outcome metrics (Ariyurek, Surer, and Betin-Can 2023), or trace similarity (Ingram et al. 2023), but none have incorporated structured human feedback to assess whether behaviors are actually *perceived* as consistent with the intended personas. While quantitative approaches can confirm that agents diverge behaviorally, they are insufficient to establish whether these numerical differences translate into perceptually meaningful and recognizable archetypes.

Believability assessment has been extensively studied in HCI and game AI, from Loyall's (1997) early requirements for personality and emotion to Gomes et al.'s (2013) metrics for interactive narrative. More recently, Guo et al. (2023) developed a standardized scale covering eight dimensions. To our knowledge, no study has applied such instruments to procedural personas. This gap matters when an agent is intended to communicate a designer-defined archetype: quantitative metrics alone cannot determine whether behavioral differences translate into perceptually meaningful distinctions.

In summary, prior work has established that quantitative metrics can distinguish persona agents (SQ1), yet leaves unanswered: (SQ2) whether these quantitatively distinct profiles are perceptually recognizable to human observers, and (SQ3) whether perceived believability mediates such recognition. Our work directly addresses this by first validating the quantitative distinctiveness of RL-trained personas (SQ1), then conducting the first structured human evaluation using Guo et al.'s (2023) believability instrument to determine whether these personas are not just numerically different, but legible to human observers (SQ2 and SQ3).

3 Game Environment

The testbed used in this study is a fixed 2D arena containing the core mechanics and interactions of a multi-objective game (Pereira 2022; Pereira, Viana, and Toledo 2024). The arena geometry—walls, obstacles, and the exit tile—remains unchanged between episodes, providing a controlled environment. However, to prevent agent overfitting, the initial positions of all objects are randomized each episode: the player starts at one of ten dedicated spawn points chosen uniformly at random, while a configurable number of enemies and treasures are randomly distributed across 34 shared spawn points. This scheme preserves a consistent arena structure while exposing agents to varied configurations of interactive objects. Figure 1 shows a screenshot of the arena with its main objects and elements.

Core Mechanics

The player character has a total of 10 health points (HP), represented by red hearts, moves freely in continuous 360-degree directions, and fires projectiles in the four cardinal directions (up, down, left, right). The arena contains three types of interactive elements:

- **Treasures**: collectible items that grant a reward when the player moves over them. Treasures have no mechanical effect other than increasing the collection counter.
- **Enemies**: non-player characters (NPC) with distinct speed, HP, damage, movement patterns, and attack types.

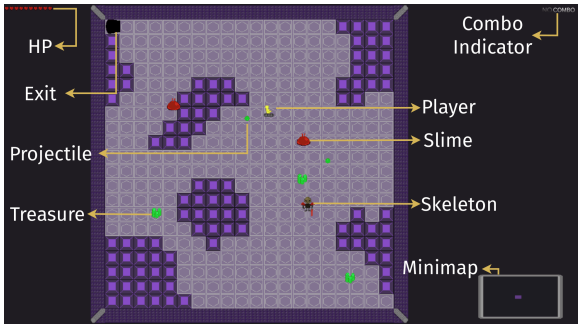


Figure 1: The game arena, showing the player character, HP bar, and projectiles; two enemies (skeleton and slime), scattered treasures, the exit door, minimap, and combo indicator.

Enemy	Speed (u/s)	HP	Movement	Damage
Skeleton	1.3	5	Seeks player	3
Slime	2.0	1	Seeks player	1
Bomber	0.5	1	Random	4
Healer	1.0	1	Random	0

Table 1: Enemy types and their parameters. Speed in world units per second (u/s).

All movement speeds are given in world units per second, where 1 unit equals the side length of one floor tile. For reference, the player character moves at 8 units/s. The current implementation includes four enemy types: two melee enemies (Skeleton and Slime) that deal damage upon contact, a ranged enemy that throws explosive projectiles (Bomber), and an enemy that heals nearby enemies instead of dealing damage (Healer). Their parameters are summarized in Table 1.

- **Exit tile**: a door-shaped cell that terminates the episode when touched.

4 Reinforcement Learning Pipeline

This section details the training pipeline for the five procedural personas, designed to produce quantitatively distinct gameplay profiles (SQ1) for subsequent human evaluation (SQ2 and SQ3).

Observation and Action Spaces

The agent receives a hybrid observation: a scalar health ratio $\in [0, 1]$ and a 360° spatial representation from a 2D Ray Perception Sensor. The sensor casts 32 rays (max length 25 units) detecting three tags: `Door`, `Enemy`, and `Treasure`. Each ray outputs a 5-float vector per the ML-Agents specification[1]: a one-hot tag encoding, a binary miss indicator, and a normalized hit fraction. This yields $32 \times 5 = 160$ spatial features plus the health scalar, totaling 161 real-valued features. No pixel inputs are used.

[1]https://docs.unity3d.com/Packages/com.unity.ml-agents@2.0/api/Unity.MLAgents.Sensors.RayPerceptionOutput.RayOutput.html

The action space is hybrid: a continuous branch for horizontal and vertical movement ($[-1, 1]$ each), and a discrete branch for shooting direction (idle, up, down, left, right).

Reward Functions and Persona Shaping

Each archetype follow the definition by Holmgård et al. (2014b) and is defined by a linear scalarization of event-based features (Abbeel and Ng 2004; Hayes et al. 2022):

$$R_p(s, a) = \sum_i w_{p,i}\, \phi_i(s, a)$$

where ϕ_i are event rewards and $w_{p,i}$ are persona-specific coefficients. We adopted the five original archetypes (Table 2), adapting weights to our continuous arena (Table 3).

Training Algorithm, Hyperparameters, and Infrastructure

All agents are trained with PPO (Schulman et al. 2017) via Unity ML-Agents (Juliani et al. 2020), using the hyperparameters in Table 4. The network has one hidden layer (128 units, ReLU, layer normalization). Training runs for 1 million steps on consumer hardware (AMD Ryzen 5 2400G, GTX 1050 Ti 4GB, 16GB RAM), with wall-clock times between 4.5–6.3 hours per persona. The complete Unity project and reproducibility instructions are available in the repository accompanying this paper.

Training Protocol

Each persona was trained with randomized initial positions of player, enemies, and treasures per episode. In the *standard configuration,* the number of enemies and treasures per episode were each drawn independently and uniformly at random from 1 to 5, and episodes terminated upon reaching the exit. Baseline, Runner, and Survivalist trained under this configuration for the full 1M steps. However, in our exploratory pilot run, the Monster Killer and Treasure Collector personas converged to an exit-only policy because the sparse combat and collection rewards were not effectively discovered. Therefore, we utilized a curriculum (Bengio et al. 2009) for sparse-reward bootstrapping:

- **Monster Killer**: For the first 500k steps, the exit was disabled and episodes terminated only after all enemies were defeated. From 500k–750k, the exit was re-enabled and episodes could terminate either by reaching the exit or by eliminating all enemies. From 750k–1M, the standard configuration was used.

Archetype	Description
Baseline	Tries to reach the exit door.
Runner	Reaches the exit as fast as possible.
Survivalist	Avoids dying and reaches the exit safely.
Monster Killer	Seeks and kills enemies, prioritizing combat.
Treasure Collector	Focuses on collecting treasures before exiting.

Table 2: Persona archetypes and behavioral descriptions.

Event (ϕ_i)	B	R	S	MK	TC
Reached exit	0.5	0.5	0.5	0.5	0.5
Step penalty	0	-0.01	0	0	0
Died	0	0	-1	0	0
Killed an enemy	0	0	0	0.5	0
Hit an enemy	0	0	0	0.5	0
Collected a treasure	0	0	0	0	1

Table 3: Persona-specific reward weights. Step penalty is applied per decision step; others are event-based.

Hyperparameter	Value
Batch size	512
Buffer size	2048
Epochs	3
Learning rate	3.0×10^{-4} (constant)
Hidden units	128
Entropy coef. (β)	0.01
PPO clip (ϵ)	0.2
GAE (λ)	0.95
Discount (γ)	0.99
Time horizon	256
Training steps	1 000 000

Table 4: PPO hyperparameters.

- **Treasure Collector**: For the first 250k steps, the number of treasures per episode was increased to 15–20 to provide a denser collection reward. From 250k–1M, the standard configuration was used.

Reward weights remained fixed throughout; only environmental parameters changed. TensorBoard monitored cumulative reward, episode length, and loss every 10k steps.

For evaluation, each final policy used greedy action selection for 100 episodes containing: a fixed player spawn not used in the training, three enemies, and three treasures. Enemy and treasure positions were randomized independently in every episode; thus, policy inference was deterministic, but the evaluation episodes were not.

5 User Study Design

This section describes the structured human evaluation designed to answer SQ2 (perceptual recognition) and SQ3 (believability association). The study pairs a forced-choice identification task with subjective rating scales, adapting the virtual-agent believability methodology of Guo et al. (2023). The design is fully within-subjects: every participant judges all five personas.

Stimuli

The stimuli are five pre-recorded gameplay clips, one per persona. Each clip was captured from the final policy of the corresponding 1M-step training run, using the evaluation-arena configuration described in Section 4. Every recording lasts exactly 30 seconds, is shot from a fixed camera, and is rendered at identical resolution and frame rate, ensuring that the agent's behavior is the sole distinguishing feature.

To avoid priming, each clip is shown under a neutral Greek-letter label rather than its archetype name. The fixed mapping is: Alpha = Baseline, Beta = Monster Killer, Gamma = Runner, Delta = Survivalist, and Omega = Treasure Collector. This mapping is withheld until after the identification and believability ratings are collected.

Participants

Participants are recruited via social media, university mailing lists, and gaming community forums. No personally identifiable information is collected. Responses that fail either attention-check item (one in the demographics one in the video block labeled Omega) are excluded before analysis. Prior to the video blocks, we collect basic demographics (age, gender, education level, gaming frequency). These items were elaborated following the literature's best practices (Silber, Roßmann, and Gummer 2022; Pei et al. 2020; Kung, Kwok, and Brown 2018).

Instrument

Data are collected via an online questionnaire (Google Forms). Before any video, an initial block gathers demographics. Next, an introductory image of the arena (similar to Figure 1) is displayed. Participants then choose one of three color-coded trails—Red, Green, or Blue. Each trail presents the same five videos in a different order (Table 5), thereby counterbalancing presentation order across the sample.

For each clip, presented one at a time, participants complete a two-stage battery. Within each stage, the order of response options is randomized.

Stage 1: blind impression (before the archetype is revealed).

- **Forced-choice identification.** Participants select which of six mutually exclusive options best describes the character's behavior: the five behavioral descriptions from Table 2 plus "*None of the options adequately describes what the character does*." This closed-set task directly tests SQ2—whether observers can map a behavior to the designer-defined archetype—while the "None" option permits rejection of all five descriptions.
- **Believability rating.** Ten items drawn from the Behavior and Awareness factors of virtual-agent believability scale (Guo, Adamo, and Mousas 2023), each rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). Two items are reverse-coded. The questionnaire was administered in Brazilian Portuguese; the English wording follows the original scale:

1. The virtual agent's behavior drew my attention.
2. I felt the virtual agent's behavior was coherent and natural.
3. I think the virtual agent's behavior was easy to understand.
4. I felt the virtual agent's behavior was appropriate to the context.
5. I felt sometimes the virtual agent behaved inappropriately.[R]

6. I felt that the virtual agent perceived the environment around it.
7. I felt that the virtual agent reacted to changes in the environment.
8. I felt that the virtual agent was aware of my presence.
9. I felt that the virtual agent was aware of the presence of other virtual agents.
10. The virtual agent was unaware of its surroundings.[R]

Stage 2: informed consistency (after the archetype is revealed). The clip's intended archetype and its description are shown, and participants rate five persona-consistency items on a 5-point Likert scale ($1 =$ strongly disagree, $5 =$ strongly agree):

1. The character's actions matched the type of character it was supposed to be.
2. The character clearly appeared to be the type it was supposed to be, and not another type.
3. The character had characteristics of its own that made it recognizable.
4. The character was clearly distinguishable from other characters.
5. I could describe its character type to another person.

These five items form an ad-hoc scale capturing two complementary aspects: whether the agent's behavior *conforms* to its intended description and whether it is *distinguishable* from other archetypes. This scale serves as a human-level analogue to quantitative differentiation metrics, directly probing perceptual coherence.

We deliberately separate the blind impression from the informed rating: the identification and believability data must be uncontaminated by knowledge of the target archetype, whereas the consistency items are meaningful only *given* that target. An optional free-text box at the end invites participants to leave comments.

The procedure has the following steps: (1) Participants access the questionnaire via a shared link. The first section presents a Free and Informed Consent Term detailing the study's purpose, voluntary participation, the right to withdraw, and data anonymization. Participants must agree to participate before proceeding; (2) They complete demographic questions; (3) They read a brief illustrated overview of the game elements: the arena, character movement, shooting, treasure collection, and the exit-door objective; (4) Participants choose one of three color-coded trails, which assigns the corresponding video order in Table 5; (5) For each clip in that order, participants watch the video (replayable if desired), then complete Stage 1 (forced-choice and believability ratings), followed by Stage 2 (reveal and consistency ratings).

The questionnaire was reviewed internally for clarity and timing; the estimated completion time is 15–20 minutes. No personally identifiable information is collected, and anonymized data are retained for at least five years. Under Brazilian CNS Resolutions No. 510/2016 and No. 674/2022, this anonymized opinion survey is exempt from ethics-committee review.

Trail	1	2	3	4	5
Red	Omega	Alpha	Delta	Beta	Gamma
Green	Alpha	Omega	Beta	Gamma	Delta
Blue	Gamma	Beta	Delta	Omega	Alpha

Table 5: Counterbalancing. Each trail fixes the presentation order of the five Greek-labeled clips.

Analysis Plan

All analyzes are performed in Python version 3.13. Responses failing either attention check are excluded first. Because the rating scales are ordinal and every participant judges all five personas, we use non-parametric, repeated-measures tests throughout.

1. **Descriptive statistics.** Per persona: forced-choice identification accuracy (proportion of participants selecting the intended archetype); mean and SD of the *Behavior* and *Awareness* believability subscales; and mean and SD of the five persona-consistency items (reverse items recoded first).
2. **Confusion structure.** A persona $\times$ chosen-label confusion matrix summarizes which archetypes are mistaken for which, and how often "None" is chosen.
3. **Inter-rater agreement.** Fleiss' κ on the forced-choice responses, overall and per persona.
4. **Internal consistency.** Cronbach's α and McDonald's ω for the Behavior subscale, the Awareness subscale, and the 5-item consistency scale.
5. **Differences across personas (ratings).** A Friedman test on the consistency score (and, secondarily, on each believability subscale) tests whether the personas are rated differently; significant omnibus results are followed by post-hoc pairwise Wilcoxon signed-rank comparisons between persona pairs, with Bonferroni correction.
6. **Differences across personas (identification).** As identification is binary per persona, we test whether accuracy differs across the five personas with Cochran's Q, followed by pairwise McNemar tests (Bonferroni-corrected).

All tests are two-tailed at $\alpha = 0.05$. Effect sizes are reported: Kendall's W for Friedman, and rank-biserial correlation for Wilcoxon. A pre-hoc (a priori) power analysis was conducted using the paired-samples t-test approximation[2], as an approximation for the conducted Wilcoxon test, with expected Cohen's $d = 0.5$, $\alpha = 0.05$, and desired power $= 0.80$. The required sample size was estimated as $N = 34$, which was exactly met by the obtained sample.

6 Results

Training and Behavioral Metrics (SQ1)

This subsection addresses **SQ1 (Quantitative Distinctiveness)**: we examine whether the trained agents produce gameplay profiles that are statistically distinguishable according to their intended archetypes.

[2]Implemented in `statsmodels.stats.power.TTestPower`

Training Dynamics Figure 2 shows the evolution of cumulative reward and episode length for all five personas over the 1 million training steps. The remaining per-persona training metrics—policy entropy, extrinsic reward, extrinsic value estimate, policy loss, and value loss—are deposited in the dataset accompanying this paper. Despite substantial differences in reward structure and convergence speed, all personas learned stable, interpretable policies that reflect their intended archetypes.

Because the Monster Killer and Treasure Collector curricula altered the environment configuration during training, Figure 2 documents each persona's learning dynamics rather than providing a controlled comparison of convergence across all five personas.

In the Unity environment, one simulation step corresponds to one visual frame, and the agent was configured to act every frame (Decision Period $= 1$); hence all step counts reported in this paper—whether from training curves or evaluation metrics—refer to game-world steps, which are also the agent's decision steps.

The **Baseline** agent converges rapidly and consistently. Cumulative reward rises from zero to an asymptote just below the theoretical ceiling of 0.5, and episode length drops from over 400 steps to roughly 15–20 steps within the first 300k steps.

The **Runner** persona faces a strong per-step penalty (-0.01) that initially drives its cumulative reward deep into negative territory (reaching -9.2 at 20k steps). However, the policy quickly learns to minimize episode length; by 400k steps it reduces steps per episode to the same 15–20 floor as the Baseline, and its cumulative reward stabilizes near -0.3. The permanent negative reward is a direct consequence of the step penalty, but the agent nonetheless achieves the fastest possible navigation.

The **Survivalist** agent starts with a slightly negative reward (-0.78) due to early deaths but eliminates them entirely within 50k steps. Episode length spikes to over 1,100 steps during an initial safety-first exploration phase, then decays smoothly to the 15–20 step floor by 400k steps. Cumulative reward climbs monotonically to 0.48—practically identical to the Baseline.

The **Treasure Collector** undergoes a stark curriculum transition at 250k steps, clearly visible as a vertical drop in both reward and episode length in Figure 2. During the high-treasure phase (15–20 treasures), cumulative reward peaks near 9.8, and episode lengthens to over 2,300 steps as the agent thoroughly explores the arena. After the transition to the standard 1–5 treasures, reward settles at ≈ 2.3 and episode length collapses to a flat floor, demonstrating that the policy generalizes smoothly to the sparser distribution.

The **Monster Killer** exhibits the most complex dynamics, reflecting its three-stage curriculum. Phase 1 (steps 0–500k, no exit) sees a gradual reward increase to 1.27 and tightly oscillating episode lengths ($\sim$1,150–1,200). At the first boundary (exit re-enabled, forced-kill still active), reward spikes to a global peak of 1.90 and episode lengths surge to 2,250, then gradually shorten to ~ 400 steps. The second boundary at 750k steps (forced-kill removed) causes another large discontinuity—episode lengths explode to 2,800 before settling into a high-variance band of 900–1,250 steps. Final reward stabilizes at ≈ 1.40.

Table 6 reports the final per-episode statistics. The Monster Killer killed the most enemies ($M = 1.66$), the Treasure Collector gathered the most treasures ($M = 2.47$), the Survivalist recorded no deaths, and the Runner completed episodes in the fewest steps ($M = 188.64$). Kruskal–Wallis tests confirmed significant differences across the five personas for all four quantitative metrics (all $p < 0.001$). Complete TensorBoard log, per-episode metrics, and the analysis code are available in the dataset accompanying this paper.

Human Evaluation – Perceptual Recognition and Believability (SQ2 and SQ3)

This subsection addresses SQ2 (Perceptual Recognition) and SQ3 (Believability Association) through forced-choice identification, consistency ratings, and believability subscales.

We recruited 37 participants via social media, professional social networks, and personal contacts. Three attention-check items (one embedded in the demographics block and two accompanying the Omega clip) excluded inattentive responses, leaving 34 participants for analysis. The final sample had a mean age of 28.0 years ($SD = 8.0$) and was 82% male (N=28), 15% female (N=5), and 3% nonbinary (N=1). Gaming frequency was daily for 17 participants (50.0%), weekly for 13 (38.2%), rarely for 3 (8.8%), and never for 1 (2.9%). All participants completed the full within-subjects questionnaire, watching the five gameplay clips (one per persona) in a counterbalanced order. The median completion time was approximately 15 minutes.

Forced-Choice Identification Accuracy (SQ2) For each clip, participants selected which archetype best matched the behavior, or chose "None of the options adequately describes what the character does." The confusion matrix (Figure 3) aggregates all 34 responses.

The Monster Killer was the most accurately identified (76%), followed by the Treasure Collector (74%), Survivalist (71%), and Runner (68%). Only 24% correctly labeled the Baseline, which was most often mistaken for the Runner (41%). The "None" option was selected infrequently (3.5% overall).

A Cochran's Q test confirmed that identification accuracy differed significantly across personas ($Q = 31.38$, $p < 0.001$). Post-hoc McNemar tests with Bonferroni correction revealed that the Baseline was identified significantly less often than every other persona (all $p < 0.01$). No other pairwise comparisons reached significance. Fleiss' $\kappa = 0.399$ indicates fair inter-rater agreement overall, with the highest agreement for the Monster Killer ($\kappa_j = 0.578$) and Treasure Collector ($\kappa_j = 0.680$), and very low agreement for the Baseline ($\kappa_j = 0.010$). These results partially support SQ2: observers can identify four of five personas above chance, but the Baseline proved indistinguishable from the Runner.

Consistency Ratings (SQ2) After the blind Stage 1, the intended persona label was revealed, and participants rated consistency on five 5-point Likert items (1 = strongly disagree, 5 = strongly agree). The items showed excellent inter-

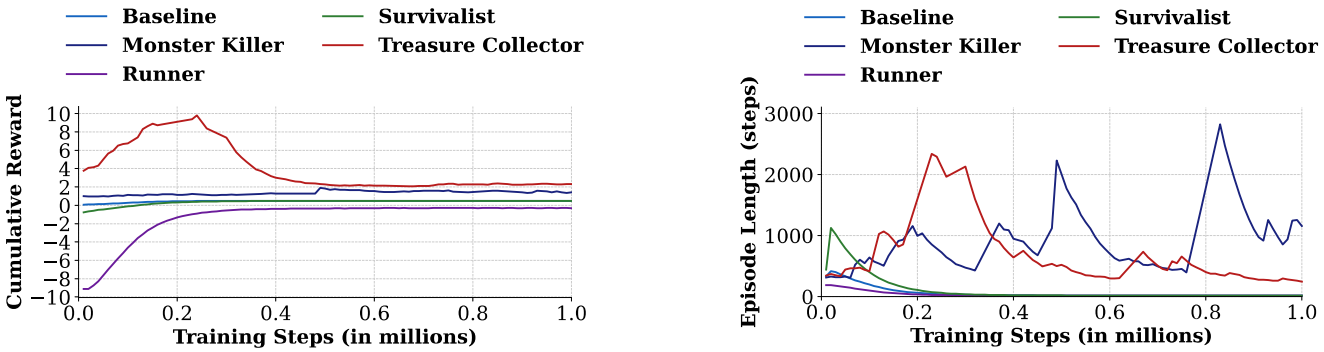


Figure 2: Training curves for all five personas over 1 million steps. Left: cumulative reward (smoothed). Right: episode length in steps.

Persona	Enemies killed	Treasures collected	Deaths	Steps	Exit rate (%)
Baseline	0.34 ± 0.54	0.42 ± 0.67	0.03 ± 0.17	195.13 ± 60.01	97
Runner	0.43 ± 0.62	0.61 ± 0.71	0.06 ± 0.24	188.64 ± 41.53	94
Survivalist	0.46 ± 0.63	0.33 ± 0.59	0.00 ± 0.00	197.73 ± 34.79	100
Monster Killer	1.66 ± 1.06	0.91 ± 0.98	0.48 ± 0.50	3888.58 ± 6706.84	52
Treasure Collector	1.21 ± 1.02	2.47 ± 0.78	0.24 ± 0.43	1151.94 ± 1461.21	76

Table 6: Behavioral metrics over 100 evaluation episodes with greedy policy execution and randomized enemy and treasure positions. Values are mean $\pm$ standard deviation; exit rate is reported as a percentage.

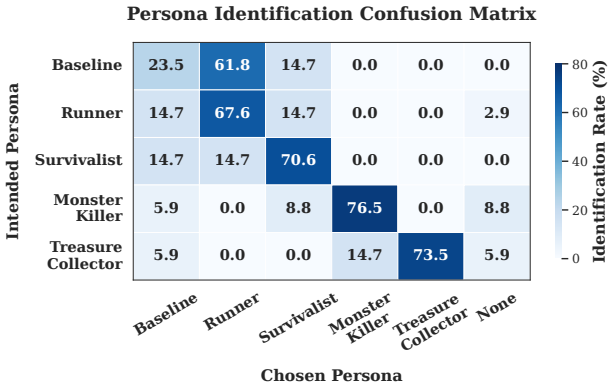


Figure 3: Confusion matrix for the forced-choice identification task. Rows show the true persona, columns the chosen label.

nal consistency (Cronbach's $\alpha = .90$, McDonald's $\omega = .90$) and were averaged into a single score.

Mean consistency scores were highest for the Runner ($M = 4.36$, $SD = 0.80$) and Survivalist ($M = 4.24$, $SD = 0.96$), followed by the Monster Killer ($M = 4.07$, $SD = 1.14$), Treasure Collector ($M = 3.76$, $SD = 1.29$), and Baseline ($M = 3.60$, $SD = 1.03$). A Friedman test indicated significant differences ($\chi^2(4) = 10.50$, $p = .033$, Kendall's $W = .077$). Post-hoc Wilcoxon signed-rank tests with Bonferroni correction showed that only the Baseline vs. Survivalist comparison reached significance ($p = .005$, rank-biserial $r = -.79$); no other pairwise difference survived correction. Consistency ratings thus reinforce the forced-choice pattern: the Baseline is perceived as significantly less consistent than the Survivalist, while other personas occupy an intermediate, statistically indistinguishable range.

Believability Subscales and SQ3 Analysis The ten believability items (Stage 1) were grouped into the Behaviour and Awareness factors of Guo et al. (2023). Internal consistency was acceptable for Behavior (Cronbach's $\alpha = .73$, $\omega = .79$) but lower for Awareness ($\alpha = .59$, $\omega = .70$); the latter should be interpreted cautiously.

Mean behavior ratings were highest for the Survivalist ($M = 5.70$, $SD = 1.14$) and Baseline ($M = 5.38$, $SD = 0.92$), followed by the Runner ($M = 4.99$, $SD = 1.28$), Treasure Collector ($M = 4.74$, $SD = 1.45$), and Monster Killer ($M = 4.55$, $SD = 1.58$). A Friedman test showed significant differences ($\chi^2(4) = 25.32$, $p < .001$, Kendall's $W = .19$). Post-hoc Wilcoxon tests with Bonferroni correction revealed that the Survivalist was rated significantly higher than the Monster Killer ($p = .002$, $r = .77$) and the Treasure Collector ($p = .001$, $r = .80$); no other comparisons survived correction. Awareness ratings did not differ significantly across personas (Friedman $\chi^2(4) = 7.01$, $p = .136$), ranging from 4.30 (Runner) to 4.84 (Survivalist).

To address SQ3, we examined the relationship between believability and recognition accuracy. Spearman's correlation across personas revealed a moderate negative association between Behavior ratings and identification accuracy ($r_s = -0.60$, $p = .28$), driven primarily by the Survivalist's high believability but moderate accuracy, and the Monster Killer's low believability but high accuracy.

Open-Ended Comments At the end of the questionnaire, participants could leave optional remarks. Several noted that the Baseline, Runner, and Survivalist were difficult to tell apart because their behaviors all involved heading directly toward the exit. One participant suggested that showing a reference clip of each archetype before the blind task might help observers calibrate expectations. Another expressed confusion between the Monster Killer and Treasure Collector: the Treasure Collector killed all enemies on its way, which made it appear more like a Monster Killer. A few participants commented that the visual design could be improved to make the personas more recognizable—for example, by giving each archetype a distinctive item or accessory—and that the color similarity among agents, treasures, enemies, and projectiles might be problematic for color blind viewers. Overall, the written feedback reinforces the quantitative finding that subtle personas (Baseline, Runner) are harder to distinguish, while also pointing to practical usability considerations for future protocol iterations.

7 Discussion

This section discusses our findings in relation to the three research questions: SQ1 (quantitative distinctiveness), SQ2 (perceptual recognition), and SQ3 (believability association). We then address limitations and implications for automated playtesting.

Answering SQ1: Quantitative Distinctiveness. The behavioral profiles in Table 6 align with the intended reward design: the Monster Killer and Treasure Collector diverge sharply on their signature metrics. Among the exit-focused personas, distinctions are subtler but still interpretable: the Runner completes episodes fastest, the Survivalist records no deaths, and the Baseline occupies an intermediate profile. The Kruskal–Wallis tests reported in the Results section confirm that these profiles are not drawn from the same distribution. Thus, SQ1 is affirmatively answered at the set level: reward-shaped RL agents produce quantitatively differentiated gameplay profiles aligned with their intended archetypes. These relative patterns are consistent with Holmgård et al. (2014b), despite our environment being a continuous arena with more complex enemy behaviors, suggesting that the reward shaping approach transfers across distinct game environments.

However, the exit-focused personas remained quantitatively close on steps and deaths, raising the question of whether their subtler differences would be perceptually recognizable (SQ2).

Answering SQ2: Perceptual Recognition. The forced-choice identification task revealed that observers correctly identified the intended archetype at rates significantly above chance (16.7%) for four of five personas: Monster Killer (76%), Treasure Collector (74%), Survivalist (71%), and Runner (68%). Only the Baseline (24%) was indistinguishable from chance, being most often mistaken for the Runner (41% of responses). Cochran's Q test confirmed significant differences across personas ($Q = 31.38$, $p < 0.001$), with post-hoc tests showing the Baseline identified significantly less often than every other persona (all $p < 0.01$). The consistency ratings reinforced this pattern: the Baseline received the lowest mean consistency score ($M = 3.60$, $SD = 1.03$), significantly lower than the Survivalist ($p = .005$), while no other pairwise differences survived correction.

SQ2 is partially answered: observers can recognize extreme personas (Monster Killer, Treasure Collector) and the Runner above chance, but the Baseline is perceptually indistinguishable from the Runner. The Survivalist, despite above-chance identification, was not rated as significantly more consistent than the Baseline once its label was revealed. Why were the exit-oriented personas so difficult to distinguish? The reward weights show that the Runner differs from the Baseline only by a per-step penalty of -0.01, and the Survivalist only by a death penalty of -1. These subtle differences produce visually similar behaviors—all three move directly toward the exit—with only fine-grained timing and cautious maneuvering distinguishing them. In a 30-second clip, these differences were too subtle for observers to decode consistently. Thus, legibility depended on whether reward shaping produced visually salient behavioral signatures: the aggressive enemy-clearing of the Monster Killer and the prolonged item-collection of the Treasure Collector were such signatures; the fine-grained timing differences among the exit-focused personas were not. This finding is consistent with Holmgård et al. (2014b), where the Survivalist also showed lower agreement with human decision traces.

Answering SQ3: Believability Association. The believability subscales revealed a surprising pattern: the Survivalist received the highest Behavior ratings ($M = 5.70$, $SD = 1.14$), significantly higher than the Monster Killer ($p = .002$) and Treasure Collector ($p = .001$), while the Monster Killer received the lowest ($M = 4.55$). Awareness ratings did not differ significantly across personas. Spearman's correlation across personas revealed a moderate negative association between Behavior ratings and identification accuracy ($r_s = -0.60$, $p = .28$), indicating that believability and recognizability are decoupled: a persona can be perceived as behaviorally coherent without being easily identifiable (Survivalist), and vice versa (Monster Killer). Extreme, utility-maximizing behavior appears mechanical and less human, whereas more subtle, survival-oriented movement reads as more natural. A plausible explanation lies in the training process: the Monster Killer and Treasure Collector required curricula to balance conflicting objectives, producing visibly noisier movement, while the exit-focused personas trained without curricula on simpler objectives and produced efficient, predictable behavior interpreted as more believable. SQ3 is thus answered negatively: perceived believability is not associated with recognition accuracy; in fact, the most recognizable personas were rated least believable.

Limitations. Several limitations should be considered. The study used a single static arena, testing only whether behaviors generalize across randomized spawns, not unseen layouts—a known challenge (Holmgård et al. 2016a). Each persona was trained with a single random seed due to computational constraints, so quantitative metrics may reflect idiosyncrasies of that particular run. Multiple training seeds are recommended for reliable DRL comparisons (Henderson et al. 2018). The "Baseline" archetype name was retained for continuity with the first procedural persona defini-

tions (Holmgård et al. 2014b), but unlike the other labels it does not describe a behavior; this ambiguity may have influenced responses, particularly given its near-zero agreement ($\kappa = 0.010$, versus 0.578 and 0.680 for Monster Killer and Treasure Collector).

The closed-set identification task measures alignment with designer-defined labels rather than eliciting participants' spontaneous categories. Although the "None" option was selected in only 6 of 170 responses (3.5%), free-text descriptions may reveal different groupings. We also did not include videos of humans instructed to enact each persona, so the believability scores lack a human-performance baseline. The convenience sample consisted primarily of members of the general public rather than professional designers or testers; player perceptions are informative for this initial validation, but they do not establish utility in production workflows. Finally, the sample size ($N = 34$) may miss smaller effects, its strong male skew (82%) limits demographic generalizability, and the Brazilian Portuguese translation the believability scale scale (Guo, Adamo, and Mousas 2023) was not formally cross-culturally validated, and the otherwise reliable Stage 2 consistency scale has not been independently validated.

Implications for Automated Playtesting. Procedural personas are intended to assist human QA staff by expanding test coverage, not to replace the contextual judgment and labor of human playtesters. Designers can deploy agents with strong, unambiguous reward profiles—Monster Killer and Treasure Collector—for testing combat or collection mechanics, as observers reliably recognize these archetypes (SQ2). For exit-focused personas, current reward weights do not produce reliably distinguishable behaviors, making them less suitable for tests requiring perceptually distinct playstyles. The believability data add nuance: the Survivalist was rated most believable yet not most identifiable, indicating that perceived naturalness alone does not guarantee legibility. Thus, developers should not assume believable movement implies a recognizable archetype. Moreover, SQ3's negative answer suggests that optimizing for believability may trade off against recognizability, an important design consideration when balancing agent naturalness and distinctiveness for playtesting. Finally, our open-source repository developed with consumer-grade hardware makes this playtesting pipeline accessible to small studios and independent researchers, providing a reproducible baseline for benchmarking improvements in persona expressiveness and perceptual alignment.

8 Conclusion

We trained five procedural persona agents in a continuous 2D arena using reward-shaped PPO and evaluated them through quantitative gameplay metrics and a within-subjects human identification study ($N = 34$). The quantitative profiles aligned with the intended archetypes, suggesting that the reward-shaping strategy can transfer to a continuous environment (SQ1). Human observers reliably identified the Monster Killer and Treasure Collector above chance, whereas the Baseline was frequently confused with the Runner, while the Survivalist—despite being identified above chance—was not perceived as significantly more consistent than the Baseline once its label was revealed (SQ2). Only the more extreme reward profiles produced visually salient, recognizable behaviors; the subtler exit-focused personas remained difficult to distinguish.

Importantly, recognition and believability did not align: the Survivalist was rated the most believable agent, while the most accurately identified personas received the lowest believability scores (SQ3). This decoupling indicates that quantitatively distinct reward profiles can produce behaviors that are either recognizable or believable, but not necessarily both. Our open-source pipeline provides a replicable baseline for advancing persona-based playtesting research.

Future Work. Several directions remain open. First, a similar human evaluation could pair each agent clip with a recording of a human player instructed to enact the same archetype, providing a baseline for believability. The current study showed that the personas can handle randomized object placements within a fixed arena; a natural next step is to test whether they generalize to procedurally generated levels with different wall configurations and spatial structures. More mechanically complex testbeds could also offer richer opportunities for persona-specific skill expression, potentially improving visual distinguishability. Implementing the developing persona framework (Ariyurek, Surer, and Betin-Can 2023) would allow researchers to assess whether dynamic personas are perceived as more realistic than static archetypes. Finally, qualitative studies with professional game developers, similar to the interview-based investigation of automated testing practices by Politowski et al. (2022), could clarify how persona-based agents might be integrated into real-world development pipelines and what usability requirements such tools would need to meet.

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